



Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S) & LIIT

JEE-MAIN

Date: 26-12-2025

Time: 09.00Am to 12.00Pm

GTM-05

Max. Marks: 300

KEY SHEET

MATHEMATICS

1	3	2	3	3	4	4	2	5	3
6	4	7	1	8	2	9	1	10	1
11	2	12	3	13	1	14	3	15	1
16	3	17	4	18	1	19	3	20	4
21	16	22	3	23	2	24	3	25	7

PHYSICS

26	2	27	3	28	1	29	2	30	1
31	1	32	1	33	2	34	4	35	3
36	4	37	4	38	2	39	4	40	1
41	4	42	1	43	1	44	4	45	1
46	1234	47	23	48	1234	49	1234	50	12

CHEMISTRY

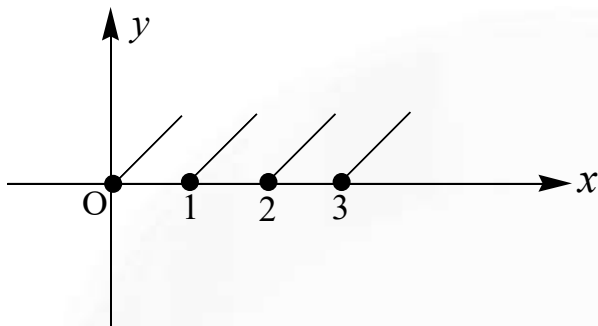
51	4	52	4	53	1	54	1	55	1
56	1	57	1	58	4	59	3	60	3
61	3	62	2	63	2	64	3	65	4
66	1	67	1	68	2	69	1	70	3
71	3	72	4	73	5	74	3	75	4



SOLUTION

MATHEMATICS

1. STATEMENT-I: Every integral point is a point of minima is true, we can observe it from the graph of $f(x) = \{x\}$



$$f(I) = 0, I \in \text{Integers}$$

$$f(I^+) = 0^+, I \in \text{Integer.}$$

$$f(I^\ominus) = I^\ominus$$

$$\therefore f(I^+) > f(I)$$

$$f(I^\ominus) > f(I)$$

$\therefore$ It is a point of minima.

STATEMENT-II: The minimum of the function occurs at median of

1, 2, 2, 3, 3, 3, 4, 4, 4, 4, ..., $\underbrace{10, 10, 10, \dots}_{10 \text{ times}}$ which is 7.

2. We know that incident ray, reflected ray and normal lie in same plane, and angle of incidence = angle of reflection.

Therefore unit vector $\hat{a}$ will be along the angle bisector of $\hat{w}$ and $-\hat{v}$,

$$\text{i.e., } \hat{a} = \frac{\hat{w} + (-\hat{v})}{|\hat{w} - \hat{v}|} \quad \dots(1)$$

[$\therefore$ Angle bisector will along a vector dividing in same ratio as the ratio of the sides forming that angle.]

$$\text{Since } |\hat{w} - \hat{v}| = OC = 2OP = 2|\hat{w}| \cos \theta = 2 \cos \theta$$

Substituting this value in equation (1) we get

$$\hat{a} = \frac{\hat{w} + (-\hat{v})}{2 \cos \theta}$$

$$\therefore \hat{w} = \hat{v} + (2 \cos \theta) \hat{a} = \hat{v} - 2(\hat{a} \cdot \hat{v}) \hat{a} \quad [\because \hat{a} \cdot \hat{v} = -\cos \theta]$$

3. STATEMENT-I : $F(x) = \int \sin^2 x dx = \frac{x}{2} - \frac{\sin(2x)}{4} + C, F(x + \pi) = \frac{x + \pi}{2} - \frac{\sin(2x)}{4} + C + \frac{\pi}{2}$

$$\therefore F(x) \neq F(x + \pi)$$

STATEMENT-II: Use mean value theorem of continuity



4. Statement-I:

$$P(E) = \frac{1}{8}; P(F) = \frac{1}{6}; P(G) = \frac{1}{4}; P(E \cap F \cap G) = \frac{1}{10}$$

$$\text{Now, } P(E^c \cap F^c \cap G^c)$$

$$= 1 - P(E \cup F \cup G) \geq 1 - \frac{13}{24} \Rightarrow P(E^c \cap F^c \cap G^c) \geq \frac{11}{24}$$

$$P(E \cup F \cup G) = P(E) + P(F) + P(G) - P(E \cap F) -$$

$$P(F \cap G) - P(G \cap E) + P(E \cap F \cap G)$$

$$= \frac{1}{8} + \frac{1}{6} + \frac{1}{4} - \sum P(E \cap F) + \frac{1}{10}$$

$$= \frac{13}{24} + \frac{1}{10} - \sum P(E \cap F)$$

$$\Rightarrow P(E \cup F \cup G) \leq \frac{13}{14}$$

Statement-II:

$$\text{Let } P(w_1) = k \text{ then } P(w_2) = \frac{k}{2} \dots P(w_n) = \frac{k}{2^n - 1}$$

$$\text{We have, } \sum_{i=1}^{\infty} P(w_i) = 1 \Rightarrow \frac{\lambda}{1 - \frac{1}{2}} = 1 \Rightarrow \lambda = \frac{1}{2}$$

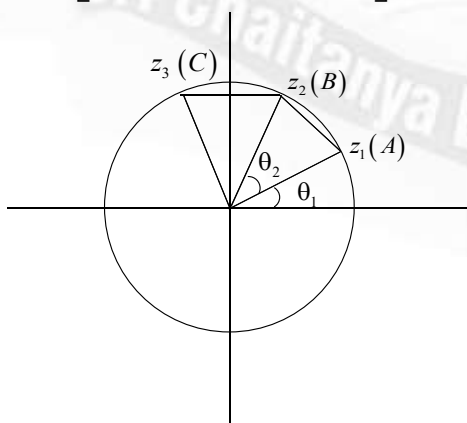
$$\sum_{i=1}^{\infty} P(w_i) = k$$

$$A = \{2K + 3\ell; k, \ell \in N\} = \{5, 7, 8, 9, 10, \dots\} \quad \text{and } B = \{w_n : n \in A\}$$

$$\therefore B = \{w_5, w_7, w_8, w_9, w_{10}, w_{11}, \dots\}$$

$$\therefore P(B) = 1 - [P(W_1) + P(W_2) + P(W_3) + P(W_4) + P(W_6)]$$

$$= 1 - \left[\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{64} \right] = 1 - \frac{32 + 16 + 8 + 4 + 1}{64} = \frac{3}{64}$$



5.

$$\text{Since } |z_1| = |z_2| = \dots = |z_{10}| = 1$$



$$\theta_2 = \text{arc}(z_1 z_2)$$

$$|z_2 - z_1| = \text{length of line AB} \leq \text{length of arc AB}$$

$$|z_3 - z_2| = \text{length of line BC} \leq \text{length of arc BC}$$

$\therefore$ Sum of length of these 10 lines $\leq$ Sum of length of arcs (i.e. 2π)

$$[\because \theta_1 + \theta_2 + \theta_3 + \dots + \theta_{10} = 2\pi]$$

$$\therefore P: |z_2 - z_1| + |z_3 - z_2| + \dots + |z_1 - z_{10}| \leq 2\pi$$

P is true.

$$\text{Now, } |z_2^2 - z_1^2| = |z_2 - z_1| |z_2 + z_1|$$

We know that

$$|z_2 + z_1| \leq |z_2| + |z_1| \leq 2$$

$$\therefore |z_2^2 - z_1^2| + |z_3^2 - z_2^2| + \dots + |z_1^2 - z_{10}^2| \leq$$

$$2(|z_2 - z_1| + |z_3 - z_2| + \dots + |z_1 - z_{10}|) \leq 2(2\pi) \Rightarrow Q \leq 4\pi$$

Q is also true.

6. Statement-I: $\lim_{x \rightarrow \infty} \frac{|\sin x|}{\sqrt{1 - \cos^2 x}} = \lim_{x \rightarrow \infty} \frac{|\sin x|}{|\sin x|} =$ Does not exist because when $x \rightarrow \infty$ $\sin x$ can be zero, limit doesn't exist.

Statement-II:

$$(3^{1/2} - 3^{1/3})^n < (3^{1/2} - 3^{1/3})(3^{1/2} - 3^{1/5}) \dots \dots \dots \left(3^{1/2} - 3^{\frac{1}{2n+1}}\right) < (3^{1/2} - 3^{1/2n+1})^n$$

$$(3^{1/2} - 3^{1/3})^n < L < \left(2^{1/2} - 2^{\frac{1}{2n+1}}\right)^n$$

$$\text{Now, } \lim_{n \rightarrow \infty} (3^{1/2} - 3^{1/3})^n = 0$$

$$\lim_{n \rightarrow \infty} (3^{1/2} - 3^{1/2n+1})^n = 0$$

By sandwich theorem we get $\lim_{n \rightarrow \infty} L = 0$.

7. $\sin \frac{1}{x}$ is bounded but oscillating

$$\tan \frac{1}{x} \text{ is unbounded.}$$

Hence $f(x)$ is continuous & $g(x)$ is discontinuous

8. STATEMENT-I:



$$A^2 B^2 - B^2 A^2 = P \quad -B = B^T$$

$$P^T = (A^2 B^2)^T - (B^2 A^2)^T \quad A = A^T$$

$$= (B^T)^2 (A^T)^2 - (A^T)^2 (B^T)^2$$

$$= B^2 A^2 - A^2 B^2$$

$$P^T = -P \quad P \text{ is a skew symmetric matrix of even order}$$

$|P|$ is perfect square. $PX = O$ has unique or infinite solutions.

Statement-II: $x = \pm a, y = \pm b, c = \pm a$.

So, system of equation has 8 solutions.

9. Using binomial expansion, we get an irrational number.

$$10. \quad f(x) - g(x) = \begin{cases} -x; & x = \text{rational} \\ x; & x = \text{irrational} \end{cases}$$

For every rational no $\rightarrow (p) : (f - g)(-p) = p$

For every irrational no $\rightarrow (q) : (f - g)(+q) = q$

$\therefore (f - g)(x)$ is one-one & onto

$$11. \quad \sin^{-1} x \text{ is increasing function: } \alpha = 3 \sin^{-1} \left(\frac{6}{11} \right) > 3 \sin^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{2}$$

$$\cos^{-1} x \text{ is decreasing function: } \beta = 3 \cos^{-1} \left(\frac{4}{9} \right) > 3 \cos^{-1} \left(\frac{1}{2} \right) = \pi$$

$$\therefore \alpha > \frac{\pi}{2}, \beta > \pi \therefore \cos \beta < 0$$

12. 'θ' angle made with 'x' & 'y' axes

'α' be the angle made with 'z' axis

$$2 \cos^2 \theta + \cos^2 \alpha = 1$$

$$2 \cos^2 \theta = \sin^2 \alpha$$

$$\cos^2 \theta = \frac{\sin^2 \alpha}{2} \quad \left\{ \sin^2 \alpha \in [0, 1] \right\}$$

$$\cos^2 \theta \in \left(0, \frac{1}{2} \right)$$

$$\cos \theta \in \left(0, \frac{1}{\sqrt{2}} \right] \quad \left(\text{since } \theta \in \left(0, \frac{\pi}{2} \right] \right)$$

$$13. \quad \vec{r} = (2l+1)\hat{i} + (l)\hat{k} - (1)\hat{j}$$

$$\vec{r} = (m+2)\hat{i} + (m-1)\hat{j} - (m)\hat{k}$$

Since we need to find intersection.

$$m-1 = -1 \quad (\text{from } \hat{j}) \quad \text{But, } l \neq -m$$

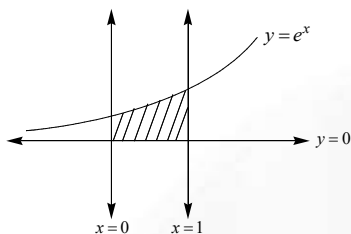


$$m = 0,$$

$$2 = 2l + 1 \quad l = \frac{1}{2} \left[\text{from } \hat{i} \right)$$

But with these it doesn't so satisfy along $\hat{k}$ direction.

14.



$$S = \int_0^1 e^x dx = e - 1$$

$$\therefore S = e - 1$$

$$S > \frac{1}{e}$$

$$S > 1 - \frac{1}{e}$$

$$S \leq \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}} \left[1 - \frac{1}{\sqrt{2}} \right]$$

$$\text{But, } S > \frac{1}{4} \left(1 + \frac{1}{\sqrt{e}} \right)$$

15. $\frac{dy}{dx} + 2y = 1 \quad x \in [0, 1]$

$$\frac{dy}{1-2y} = dx$$

$$\int_0^{y(1)} \frac{dy}{1-2y} = \int_0^2 dx$$

$$\frac{-1}{2} \ln(1-2(y(1))) = 1$$

$$1-2(y(1)) = e^{-2}$$

$$2(y(1)) = 1 - e^{-2}$$

$$y(1) = \frac{1 - e^{-2}}{2}$$

$$\frac{dy}{dx} + 2y = 0$$



$$\int_{y(1)}^{y(3/2)} \frac{dy}{-2y} = \int_1^{3/2} dx$$

$$\frac{-1}{2} \ln \left(\frac{y(3/2)}{y(1)} \right) = \frac{1}{2}$$

$$\frac{y(3/2)}{y(1)} = e^{-1}$$

$$y(3/2) = \frac{y(1)}{e} = \frac{e^2 - 1}{2e^3}$$

$$y\left(\frac{3}{2}\right) = \frac{e^2 - 1}{2e^3}$$

16. $A = (a, 0) \quad B = (-a, 0) \quad \forall a \in (-\infty, 0)$

$$PA = n PB$$

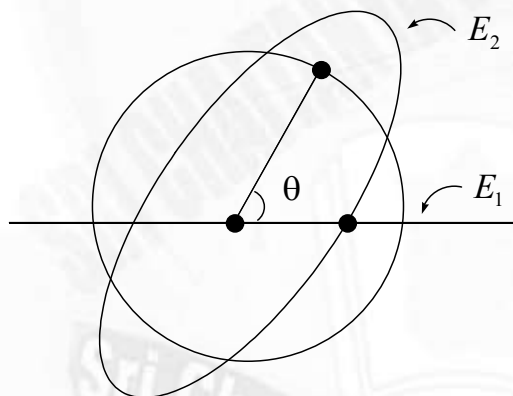
$$\sqrt{(x-a)^2 + y^2} = n \sqrt{(x+a)^2 + y^2}$$

$$x^2 + a^2 - 2ax + y^2 = nx^2 + na^2 + 2nax + ny^2$$

$$(n-1)x^2 + (n-1)y^2 + 2ax(n+1) + na^2 = 0$$

If $0 < n < 1$ then locus of p is a circle & A lies inside & B lies outside the circle

17.



$$(x, y) = (ae' \cos \theta, ae' \sin \theta)$$

$$\frac{x^2}{a^2} + \frac{y^2}{a^2(1-e^2)} = 1 \quad \text{as foci of } E_2 \text{ are on } E_1$$

$$\frac{a^2(e'^2 \cos^2 \theta)}{a^2} + \frac{a'^2(e'^2 \sin^2 \theta)}{a'^2(1-e^2)} = 1$$

$$e'^2 \cos^2 \theta + \frac{e'^2(1-\cos^2 \theta)}{1-e^2} = 1$$

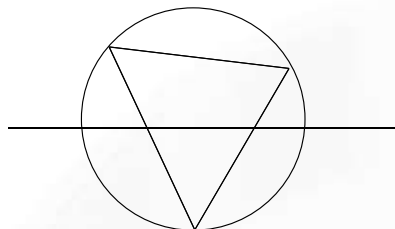
$$\cos^2 \theta (x - e^2 - 1) + 1 = \frac{1}{e'^2} - \frac{e^2}{e'^2}$$



$$(-e^2)(\cos^2 \theta) = \frac{1 - e^2 - e'^2}{e'^2}$$

$$\cos^2 \theta = \frac{e^2 + e'^2 - 1}{e^2 e'^2} \Rightarrow \cos \theta = \sqrt{\frac{e^2 + e'^2 - 1}{e^2 e'^2}} \Rightarrow \cos \theta = \sqrt{\frac{e^2 + e'^2 - 1}{e_1 e^2}}$$

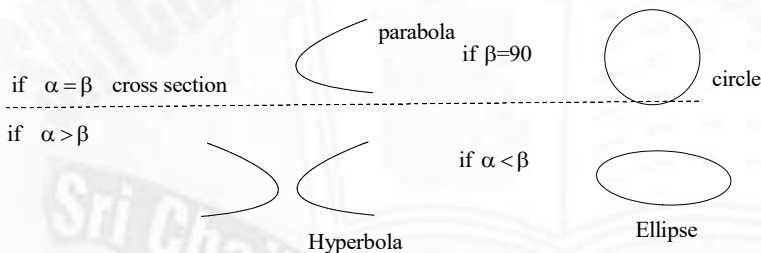
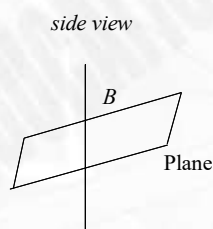
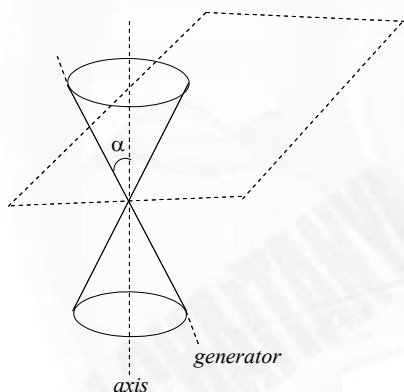
Statement-II:



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Its max area of triangle inscribed in the ellipse is $\frac{3\sqrt{3}}{4}ab$

18.



$\therefore$ Opt (1)

19. Statement-I: $p(x) = 0$ has purely imaginary roots.

then $p(p(x)) = 0$ has complex number roots which are not purely imaginary by definition

Statement-II: $S_n = \frac{n}{2}(2a + (n-1)d)$

For any progression

$S_n = 0$ has one root is 0

But 2 non-zero real & distinct roots are never possible.

20. The resulting standard deviation will be $\pm$ cs.



∴ As standard deviation is always non-negative, if we take negative constant 'c', it will be $-cs$, and if we take positive constant 'c', it will be $+cs$

21. Minimum number of elements in $S = 2^{n(A)} = 2^4 = 16$

22. i) If $a, b, c = 0$, we get an identity $0=0$ independent of x, y . Hence, as all $x, y \in R$ satisfy the equation, it represents the x - y coordinate plane.

ii) $b = 0; c \neq 0 \Rightarrow ax = -c \Rightarrow x = \frac{-c}{a}$ (st. line parallel to y -axis)

iii) $a = 0, b, c \neq 0 \Rightarrow by = -c \Rightarrow y = \frac{-c}{b}$ (st. line parallel to x -axis)

iv) If a, b, c all are not equal to zero, we get a straight line $ax + by + c = 0$ not passing through origin.

23. i) Number of obtuse angled triangles is equal to if $(2k+1) \times {}^{k-1}C_2$ if $n = 2k+1$

ii) Number of equilateral triangles = $\frac{n}{3}$, where 'n' is multiple of 3.

iii) Number of right angled triangles = $2k$, where $n=2k$

iv) Number of isosceles triangles = $\frac{n(n-1)}{2} - 2k$, where $n = 3k$ and odd

24. Correct statements: i, ii, iv

i) Trace of AA^T is sum of squares of all elements.

∴ $Tr(AA^T) \geq 0$: non negative

ii) $Adj(kA) = k^{n-1} \times Adj(A); k \in R$

iv) $(kA)^{-1} = \frac{1}{k} A^{-1}$

Wrong statement: iii

iii) Maximum no. of distinct elements in triangular matrix of order 39 is

$\frac{39^2 - 39}{2} + 39 + 1$ (including zero) = 781

25. $\sin(\alpha + \beta) + 2\sin\left(\frac{2\alpha - \beta}{2}\right)\cos\left(\frac{\beta}{2}\right) = \sqrt{2}\cos\beta + \frac{1}{\sqrt{2}}$

$\sin\alpha\cos\beta + \cos\alpha\sin\beta + \sin\alpha + \sin(\alpha - \beta) = \sqrt{2}\cos\beta + \frac{1}{\sqrt{2}}$

$2\sin\alpha\cos\beta + \sin\alpha = \sqrt{2}\cos\beta + \frac{1}{\sqrt{2}} \quad \sqrt{2}\sin\alpha(2\cos\beta + 1) = (2\cos\beta + 1)$

$\sin\alpha = \frac{1}{\sqrt{2}} \text{ or } \cos\beta = \frac{-1}{2}$

$\alpha = \frac{\pi}{4}, \frac{3\pi}{4} \text{ or } \beta = \frac{2\pi}{3}, \frac{4\pi}{3}$

∴ $(\alpha + \beta)_{\max} = \frac{4\pi}{3} + 2\pi = \frac{10\pi}{3} = \frac{m}{n}$

∴ $m - n = 7$



PHYSICS

26. AB expands more, torque about suspension will become zero.

27. Between 2 nodes all particle vibrate in same phase

28. $W_{\text{by cells}} = \Delta U + \text{Heat}$

$$-CV_2V_1 = \left[\frac{1}{2}CV_1^2 - \frac{1}{2}C(V_1 + V_2)^2 \right] + H$$

$$-CV_2V_1 = -\frac{1}{2}CV_2^2 - 2CV_1V_2 + H$$

$$H = \frac{1}{2}CV_2^2$$

Independent of V_1

29. Instantaneous sum of potential difference $V_R = V_L + V_C$

30. By de-broglie hypothesis

$$\lambda = \frac{h}{p} \quad L = \frac{n\lambda}{2} \quad \lambda = \frac{2L}{n} \quad p = \frac{h}{\lambda} = h \frac{n}{2L} \quad p = \frac{h}{\lambda} = \frac{nh}{2L}$$

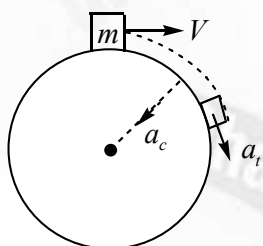
31. Backlash error is a measurement error that occurs when a mechanical system like gears or screws reverses direction. It's caused by a clearance or gap between mating components, resulting in a delay before the driven part moves in the new direction. To avoid this error, always rotate the instrument or screw in the same direction to take a measurement and only reverse direction slowly and carefully if needed.

32. $V_0 = \sqrt{\frac{GM}{r}}; U = -KE = \frac{E}{2} \quad E = -\frac{GMm}{2r}$

33. $T = \left[\pi \sqrt{\frac{K}{m}} + \frac{\ell}{v} \right] \times 2 \Rightarrow v \uparrow \Rightarrow T \downarrow$

34. Acceleration of lower most point is $\omega^2 r$ upward.

35.



up to when block have contact with sphere its net

$a_{\text{net}} = \sqrt{a_t^2 + a_c^2}$ increases as $v \uparrow$ after that only acceleration of block will be acceleration due to gravity

36. We knew that

$S_v = \text{Voltage sensitivity}$

$G = \text{Resistance of Galvanometer}$

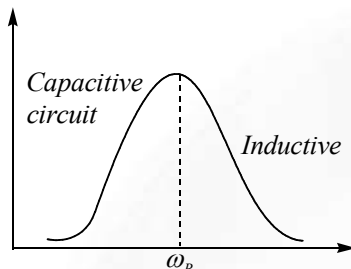
$$S_v = \frac{\theta}{V} = \frac{\theta}{IG} = \frac{Si}{G}$$



$$Sv_f = \frac{Si_f}{G_f} = \frac{\frac{3}{2} \cdot Si}{2G} = \frac{3}{4} \cdot \frac{Si}{G} = \frac{3}{4} Sv_i = \left(1 - \frac{1}{4}\right) Sv_i$$

$\therefore$ It decreased by a factor of $\frac{1}{4}$

37.



$$\omega_R = \frac{1}{\sqrt{LC}} \quad \omega_2 < \omega_R \text{ from diagram}$$

Circuit will behave as capacitance circuit

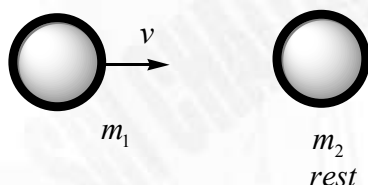
$\therefore$ Current leads voltage

38. Diamagnetic materials create an induced magnetic field in a direction opposite to an externally applied magnetic field.

Paramagnetic materials create an induced magnetic field in a direction parallel to an externally applied magnetic field.

So, these two cylinders will repel when they are placed such that their axis is common and attract when they are placed side by side.

39.

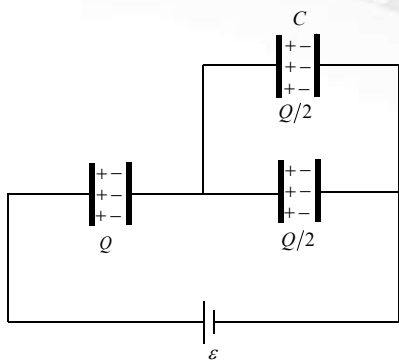


The two nuclei travel in perpendicular directions relative to each other when

$$m_2 = \frac{m_1}{e} \quad e < 1 \quad m_2 > m_1$$

40. $\vec{E} = \vec{B} \times \vec{v}$

41. Initial charge distribution is shown in figure. $Q = \frac{2C}{3} \varepsilon$





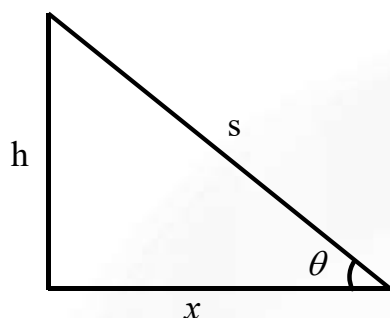
Capacitance of A increases charge increases

42. A-R, B-R, C-S, D-Q

43. Under electrostatic conditions all conductors are equipotential

No work is done by field when a charge is moved on the surface of conductor

44.



$x \rightarrow \text{fixed}$

$\theta, s, h \rightarrow \text{vary}$

$f = \mu mg \cos \theta \rightarrow \text{vary}$

45. $\gamma_{H_2} = \frac{C_p}{C_v} = 1 + \frac{2}{f} = 1 + \frac{2}{5} = \frac{7}{5} = 1.4$

$\gamma_{He} = \frac{C_p}{C_v} = 1 + \frac{2}{f} = 1 + \frac{2}{3} = \frac{5}{3} = 1.67$

$M_{H_2} = 2u$

$M_{He} = 4u$

46. Viscous forces are non-conservative

47. Lenz law

48. $\vec{F} = q(\vec{V} \times \vec{B})$

$\vec{F} \cdot \vec{V} = 0$

49. Lens displacement method

$D \geq 4f$

$f = \frac{D^2 - d^2}{4D}$

50. $Y = AB + \overline{A}\overline{B} = \overline{A \times ORB}$



CHEMISTRY

51. Based on n -factors

52. (A) The number of radial nodes in the 3p orbital is 1 For 3p: $n = 3, l = 1$ Radial nodes = $3 - 1 - 1 = 1$

(B) The 2s and 2p orbitals have identical energy in H but different radial distribution Hydrogen is a single-electron species, so all orbitals with the same n have same energy. But s and p orbitals have different shapes and radial distribution.

(C) The probability of finding the electron at the nucleus in a p-orbital is zero For p-orbitals, $l = 1 \rightarrow$ angular wave function contains $\sin\theta$, which is 0 at $r = 0$. Probability density = 0 at nucleus.

(D) The 4s orbital has three radial nodes For 4s: $n = 4, l = 0$ Radial nodes = $4 - 0 - 1 = 3$

53. Bond order $O_2^{2-} < O_2^- < O_2 < O_2^+$

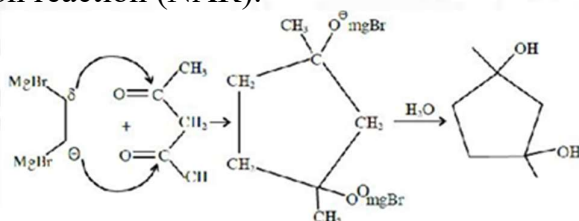
Ion/molecule	Number of e^- in BMO	Number of e^- in ABMO	Bond order
O_2^+	10	5	2.5
O_2	10	6	2
O_2^-	10	7	1.5
O_2^{2-}	10	8	1

54. MP

H_2O	273 K
H_2S	188 K
H_2Se	208 K
H_2Te	222 K

55. Both statement I & statement II are correct.

56. Although Acetyl Acetone predominantly gives Acid base reaction with G.R due to Active methylene group but according to given option ans. should be based on nucleophilic addition reaction (NAR).



57. The formation of SO_3 is exothermic and thus at high temperature backward reaction will proceed to give lower percentages yield.

58. A & B $\rightarrow$ enantiomers

A & C $\rightarrow$ diastereomers

59. BrF_5, ClF_5, IF_5

60. (a) Maltose is reducing sugar and gives silver mirror with tollen's reagent.

(b) Sucrose, starch and cellulose are non reducing sugar.

(c) Starch and cellulose are biopolymers.

(d) Sucrose has α, β (1,2) - glycosidic linkage.



(e) Starch and maltose have α -(1,4)-glycosidic linkage.

(f) Cellulose has β -(1,4)-glycosidic linkage.



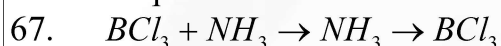
62. Both ones group -II anion i.e., do not react with dil. H_2SO_4 .

63. Assertion is correct. Reaction of ethyl bromide with sodium phenoxide yields ethyl phenyl ether.

64. Uranium - $[Rn] 5f^3, 6d^1, 7s^2$

65. Green coating on bronze is due to corrosion of Cu. Mg sacrifices itself to prevent corrosion of Fe.

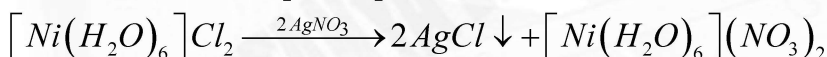
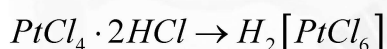
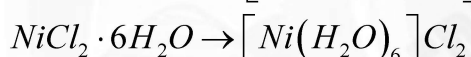
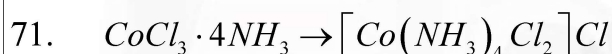
66. Energy wise distribution of molecules in a gas sample is an exponential function of temperature $e^{-E_a/RT}$ so is doubled.



68. For acidic buffer concentration of WA > SB

69. Lanthanoid Contraction

70. Reaction 'a' follows Markownikoff's rule



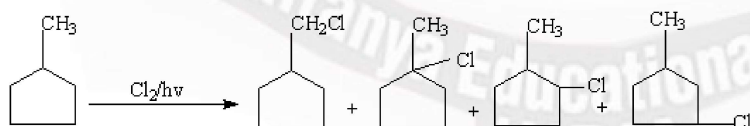
$\mu = \sqrt{2(2+2)} \text{ B.M} = 2.84 \text{ BM} \approx 3$

72. 1° and 2° amines can be distinguished using Azo dye test, Hofmann's mustard oil reaction, Carbyl amine test and Hinsberg reagent.

73. i) Boiling Point iii) Specific Gravity v) Density vii) Viscosity ix) Specific heat capacity

74. (i) Cyclopentadienyl anion
(ii) Cycloheptatrienyl cation (Tropylium ion)
(iii) Cyclopropenyl cation

75.



Free radical mechanism major